

CURRENCY COMPETITION, INTERNATIONAL INSTITUTIONS AND NETWORK TOPOLOGY

Jiri Nohejl, George Mason University

HISTORY OF INTERNATIONAL MONETARY INSTITUTIONS

classical gold standard





bancor vs dollar standard
Special Drawing Rights

floating exchange rates



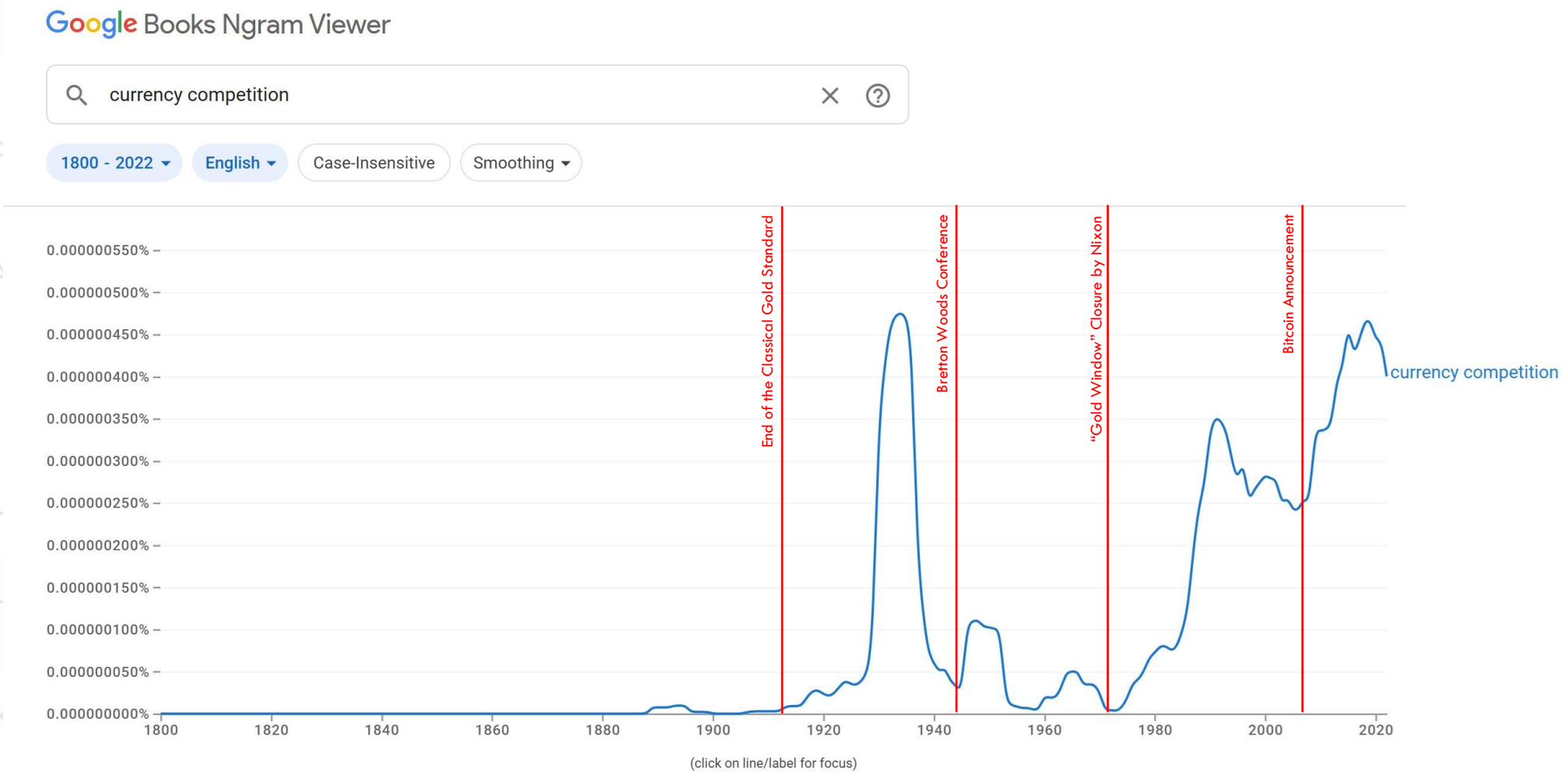


digital money challengers

CURRENCY COMPETITION IN LITERATURE

- Modern interest in currency competition started with Hayek's *The Denationalisation of Money* (1976)
- Dowd & Greenway (1993) *Currency Competition, Network Externalities and Switching Costs: Towards an Alternative View of Optimum Currency Areas* introduced formal currency competition model
- Hayek *The Future Unit of Value* debates Friedman *Currency Competition: A Sceptical View* in *Currency Competition and Monetary Union* (1984, Salin, ed.)
- Enders (2009) *Currency Competition: A Hayekian Perspective on International Monetary Integration* argues that currency competition ideal is realized by international financialization
- Luther (2016) *Cryptocurrencies, Network Effects, and Switching Costs* extends Dowd & Greenway to Bitcoin and other cryptocurrencies

CURRENCY COMPETITION IN LITERATURE



CURRENCY COMPETITION MODEL

- Model used in Down & Greenway (1993) and Luther (2016) emphasize network effects and switching costs when new money challenges incumbent money

$$u(T) = (a+bn)\int_{T}^{\infty} e^{-r(t-T)}\,dt = (a+bn)/r,$$

- Parameters:
  - a … network-independent benefits of using the currency
  - b … network-related benefit to a user [depends on the size of the network]
  - n … logarithm of size of the network (ln N) [diminishing returns to network size]
  - c … network-independent benefits for new money
  - d … network-related benefit for new money
  - s … switching costs

CRYPTOCURRENCY CHALLENGE TO STATUS QUO

- Bitcoin was introduced by Satoshi Nakamoto (2008) following financial crisis
  - to solve a "copy-paste problem" in computer science
  - to challenge incumbent money by providing digital peer-to-peer cash
- Time inconsistent launch where "money supply" expand regardless of "money demand"
- Many ICOs (Initial Coin Offerings) soon entered the competition
- Cryptocurrency competition of various "blockchain protocols"
  - Proof of Work vs Proof of Stake
  - money supply mechanisms
  - etc.
- Largely ignored by *active* regulation and labeled as "digital assets" not "money"
- **Reaction to Libra Association announcement in 2019 was a rare exception!**

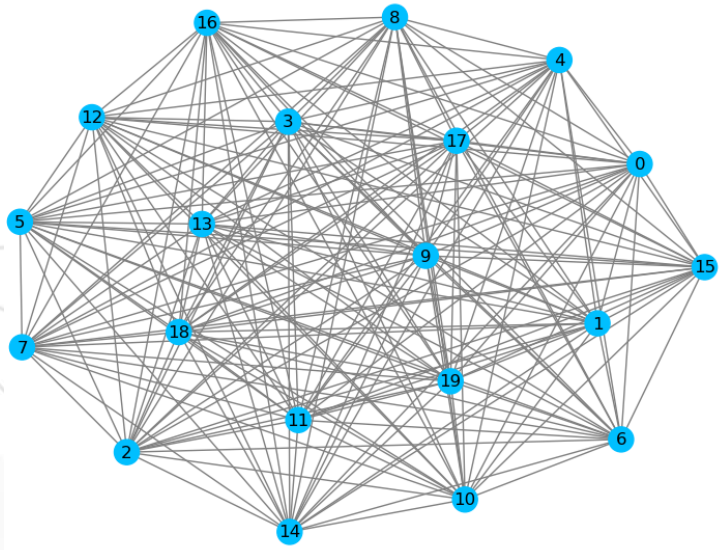
INVERSE CURRENCY COMPETITION MODEL

- The Dowd & Greenway (1993) is developed from the perspective of challenger money and respective microeconomic analysis of decisions of individual agents
- The reciprocal question is: **When the incumbent is incentivized to act?**
- Inverse model can identify conditions under which challenger can either lower switching costs or augment network effects
- Currency design can differ not only in parameters c and *dn* but also the implementation strategy
- Traditionally there has been identify two possibilities (a) government support and (b) hyperinflation of successful switching (Luther, 2016)
- **Here I propose alternative in analytical access to network topology**

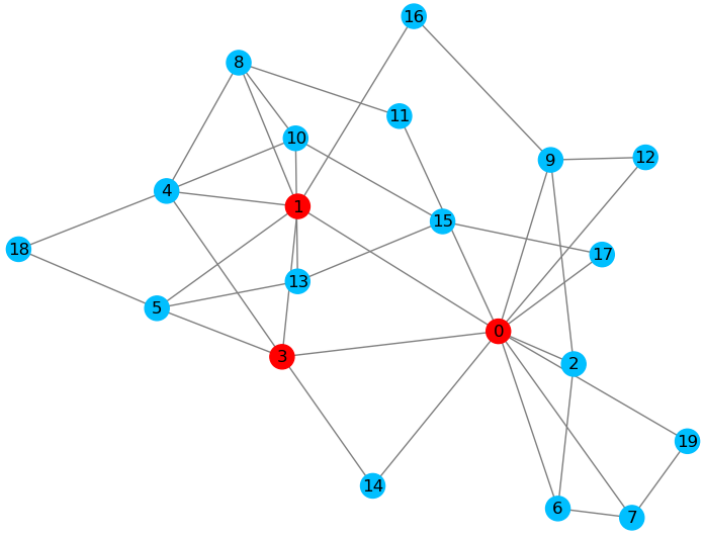
NETWORK TOPOLOGY

- Network effects are important in currency competition
- Network topology and betweenness centrality matter for analysis

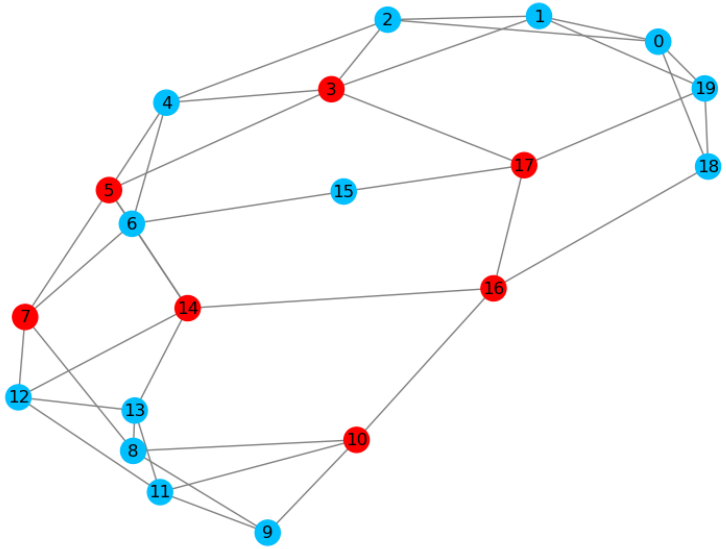
complete graph



preferential attachment graph



small-world graph



TOPOLOGY ENTERS CURRENCY COMPETITION

- Bitcoin grows its transaction network organically by adding nodes as they accept it (e.g. buying two pizzas with 10,000 bitcoins)
- Lightning Network can be used to assess network topology, but only of those already created channels (Lin et. al, 2020)

- Analytical access, utilizing computational and potentially AI methods, to an existing network can identify high betweenness centrality node (ie. hubs, bridges, …)
- Switching costs can be subsidized at identified nodes
- Currency network (or multiple layers of networks) is a key aspect of monetary institutions

PRIVATE SDR AS DYNAMIC CURRENCY BOARD

- Libra currency "backed by" by basket of currencies and other assets brings similarities with SDR (or Keynes's theoretical bancor) into perspective
- Arguments from IMF to elevate **SDRs** to "**international payment system**" (Gagnon, 2011) or "**digital retail currency**" (Coats, 2019) and arguments of **SDRs superiority** (Ocampo, 2019) or that **basket of national state currencies defeating the purpose for private currency** (Mersch, 2019)
- Libras commitment to **not "introducing inflation into the system"** conflicts with the fixed basket as **rebalancing** and revaluing policy and **active management** based on inflation expectations might be necessary
- Not only dynamic currency board doesn't defeat the purpose for private currency, it introduce the **challenge** not only to existent monies but **to prevailing international monetary institutions**

REGULATION-DRIVEN CHANGE TO PRIVATE CURRENCY

- Regulatory pressure led to:
  - diminishing size of Libra Association
  - key changes in the project
- Independent **unit of account** Libra would become a digital **wrapper** Diem over national currencies (USD-Diem, EUR-Diem, etc.)
- The required changes reflect both:
  - accuracy of the model which incorporates incumbent actions and network topology of adoption
  - absence of Hayekian competition in currency which allow private issuers of money
- Regulation selectively target project width viable design to overcome network effects and switching costs in transition from incumbent money to new money

IS CURRENCY COMPETITION FREE?

- Enders (2009) argues that Hayekian vision does not need private issuers of currency if the international competition of national currencies is open and not regulatory restricted.
- Hayek, however, argued in favor of free currency competition:

> "In advocating the freedom of offering money to the public, I am of the opinion that **everybody should be free to offer money** of differing denominations to the public. It would be the **public** which would **ultimately decide which of these monies would become generally accepted.**"
>
> Friedrich A. Hayek: *The Future Unit of Value* (1980)

- There is mostly negligible regulation of various "digital assets" because as the currency competition model suggest there is low probability of successful challenge in offering new money
- Libra Association, unlike most other ICOs, was prohibited from the competition because it offered potential competition over the standard not solely within the standard

CONCLUSION

- Evolving digital technologies give private digital currencies advantage in currency competition reflected in:
  - increasing regulation of challengers, ie. payment systems vs monetary standards
  - development of competitive public options, ie. CBDC
  - increased scrutiny of technological firms, ie. algorithm transparency

---

- Modeling network effects and switching costs in currency competition need to incorporate network topology:
  - development of agent-based models shifting focus from complete graphs to topology
  - understanding currency competition as both money quality and technological innovation
  - implication for both CBDC and FBDC (Free Banking Digital Currencies)


THANK YOU FOR ATTENTION!